

Introduction

The robot (e-puck (Cyberotics, 2019)) is simulated in different environments.

Goals:

- Task 1 (BBR):** robot follows a line while avoiding obstacles
- Task 2 (BBR):** robot moves towards light while avoiding obstacles
- Task 3 (ER)** : evolutionary system based on genetic algorithm

Task 2 – BBR Light Following (Arne)

Additional Algorithms to Task 1

- Detect Light:** detect where the light source is coming from
- Turn to Light:** turn in the direction where the light was detected
- Move to Light:** move in the direction of the light

Results & Analysis

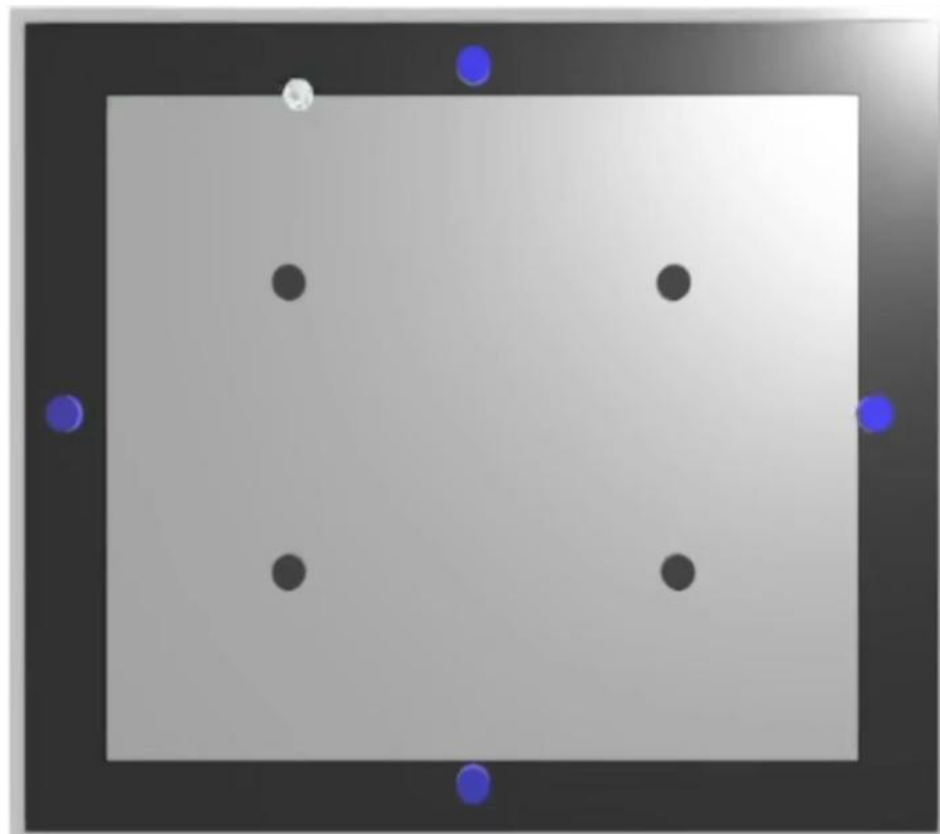


Fig 2.1 – Light Arena

Correctly detects light and moves towards it
Never hits obstacle nor wall
A bit jumpy when coming close to obstacles

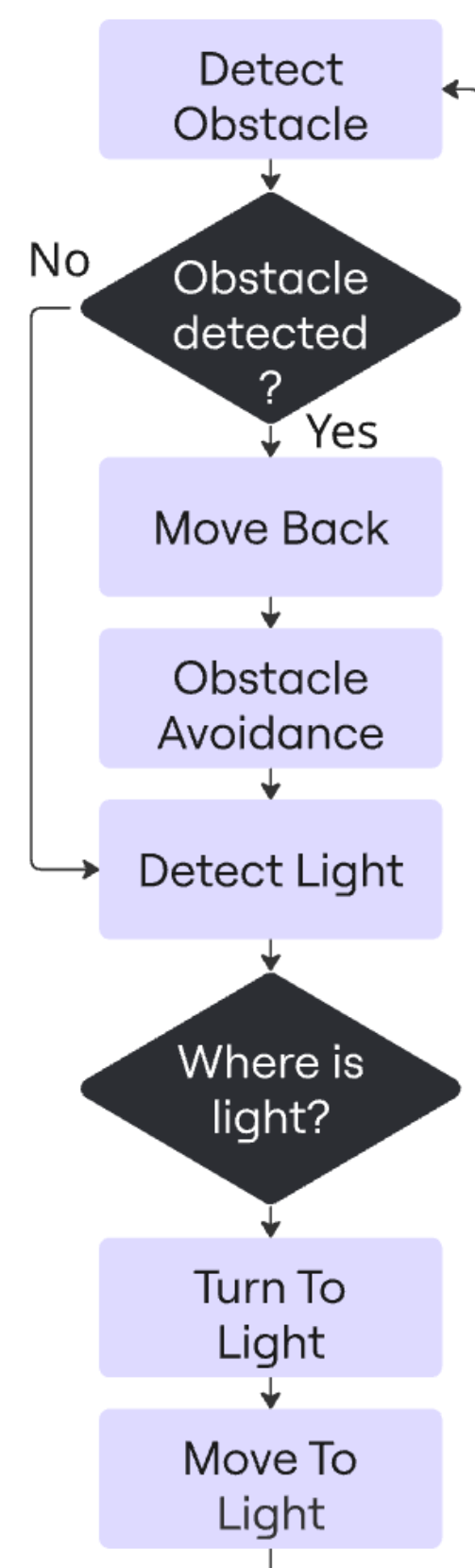


Fig 2.2 – Flow Chart

Comparative Discussion

Similarities

- Sensor driven
- Task oriented
- Requires tuning

Key Differences	BBR	ER
Control	Rule-Based	Neural Network
Transparency	Interpretable	Not Interpretable
Design Philosophy	Manual	Evolved
Determinism	Predictable	Unpredictable
Strengths	Simple, Fast	Emergent, Optimising
Weaknesses	Manual tuning, Scales poorly	Computationally costly, Fitness-sensitive

Conclusion

Key findings

- Both enable autonomous robot behaviour
- BBR = manual encoded behaviour ; ER = evolved solutions
- Choice depends on task complexity and flexibility needs

Further Improvements

BBR: Robustness against other obstacles, Richer behaviours

ER : Fitness finetuning

External Factors: Better sensors

Task 1 – BBR Line Following (Jonah)

Key Algorithms

- LineFollowing:** reads 3 ground Sensors (left/centre/right) and choses the darkest → steer accordingly
- moveBack:** fixed-time reverse
- ObstacleAvoidance:** right-wall following
- Rejoining:** right turn to intersect the line and high velocity line follower
- turn_180:** in place rotation

Results & Analysis

- follows line and does not hit any obstacles

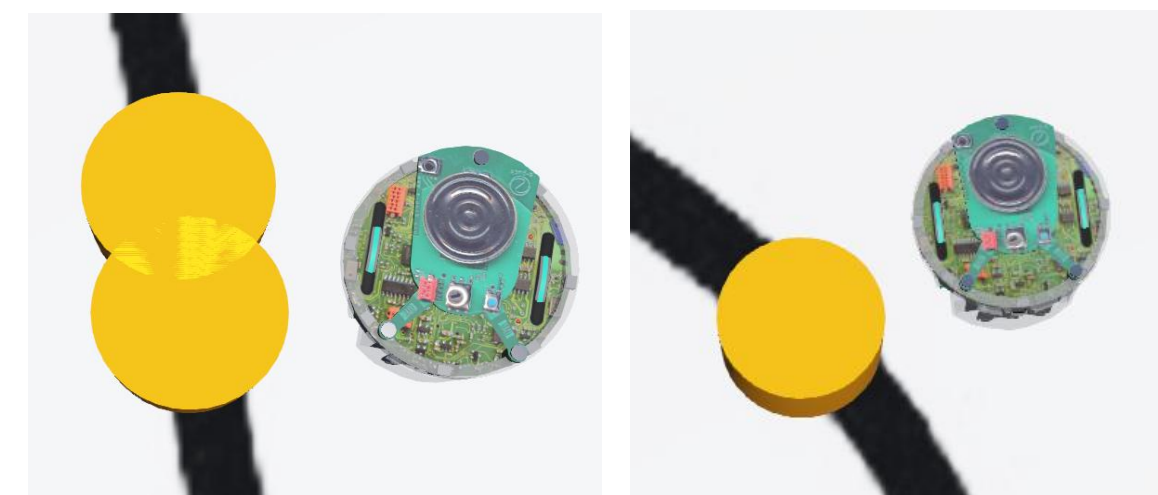


Fig 1.1 – avoiding behaviour

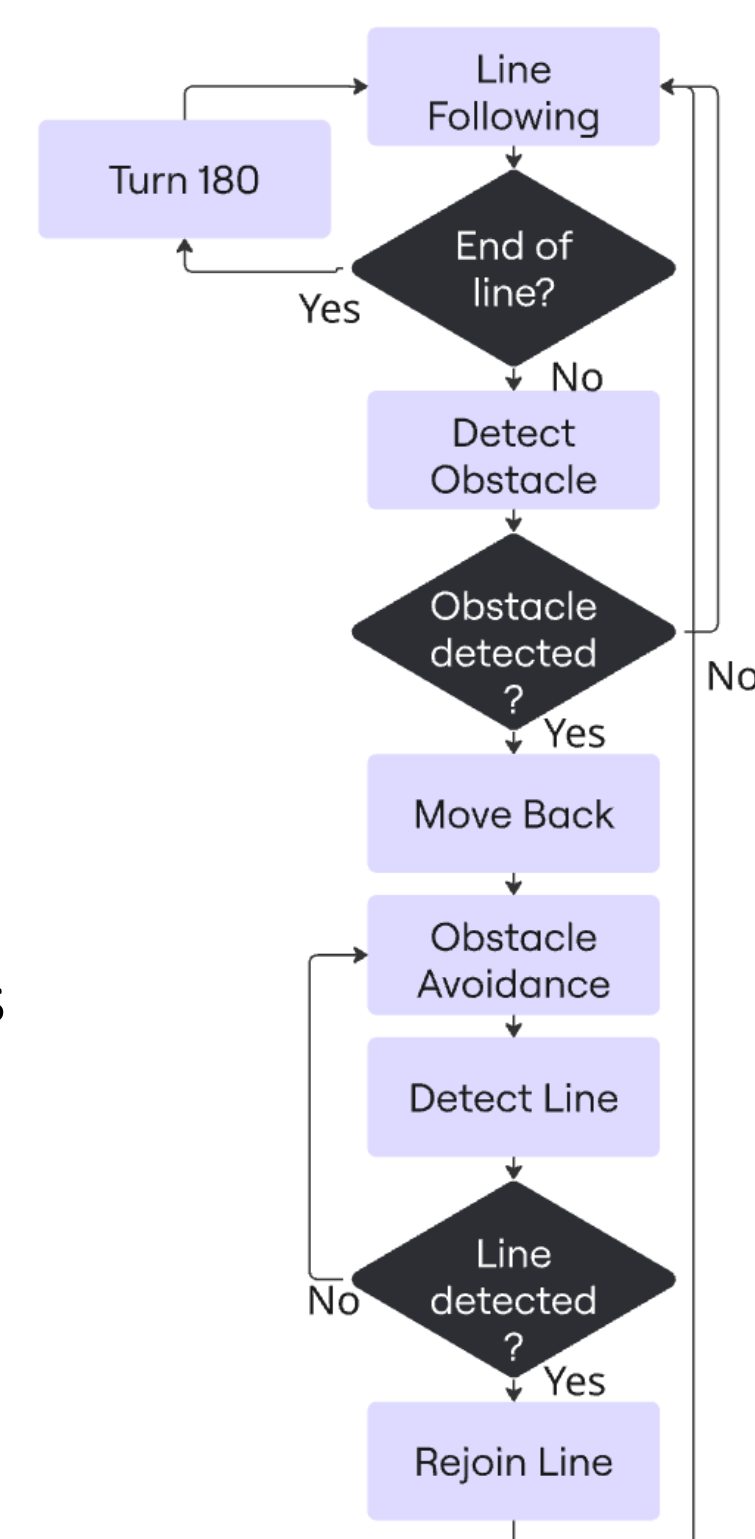


Fig 1.2 – Flow Chart

Task 3 - Evolutionary Robotics (Luke & Antoine)

Follow line -> complete a full turn avoiding obstacles. (Fig. 3.1)

Every simulation step -> fitness is calculated. (fig 3.2)

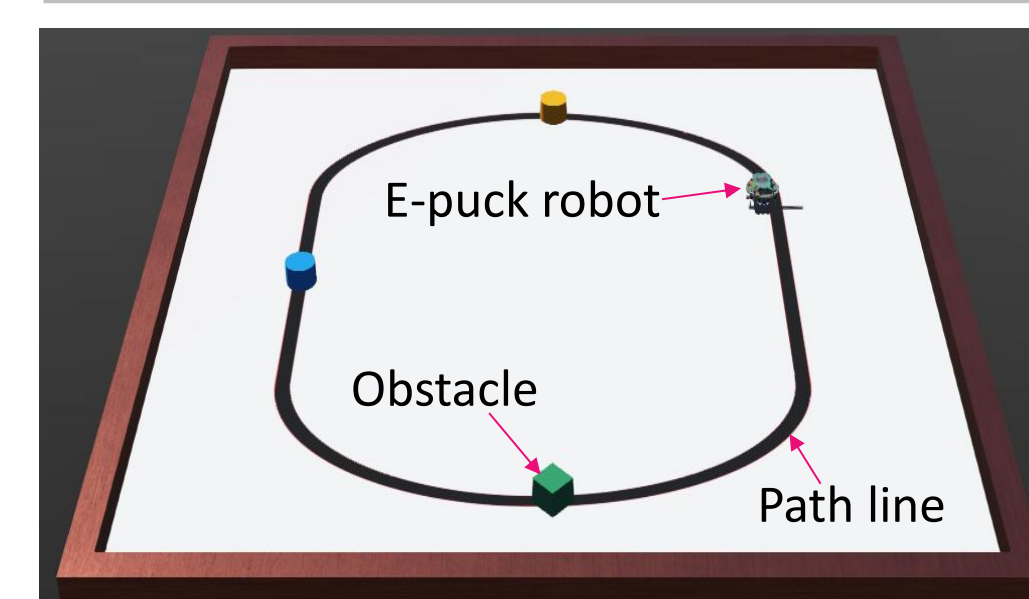


Fig 3.1 – Evolutionary Arena

Workflow

1. Understand sensors
2. Design fitness functions
3. Run trials
4. Tune genetic algorithm and fitness parameters

Fig 3.2 – Fitness

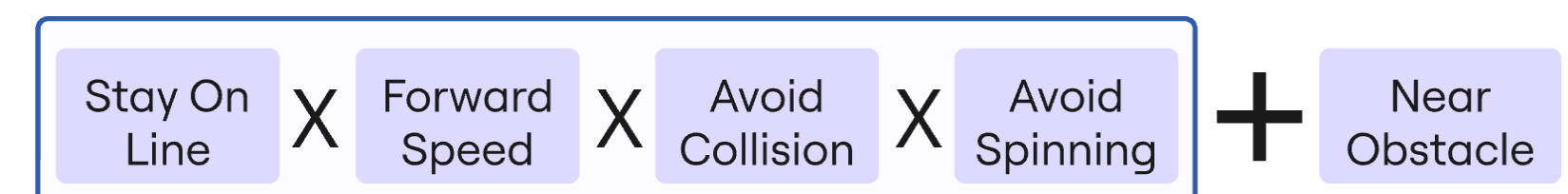
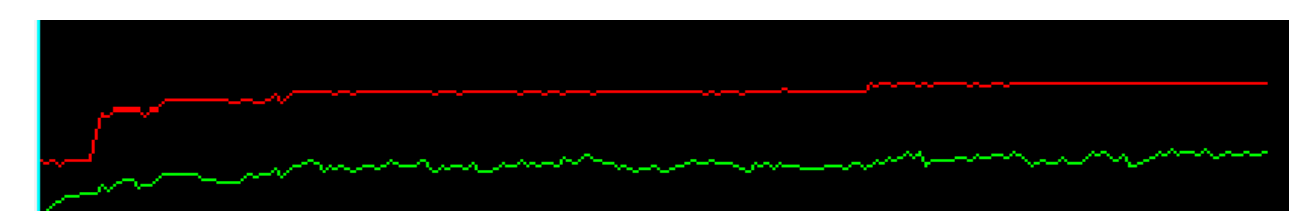


Fig 3.3 – Fitness over training: red = best green = average



Results & Analysis

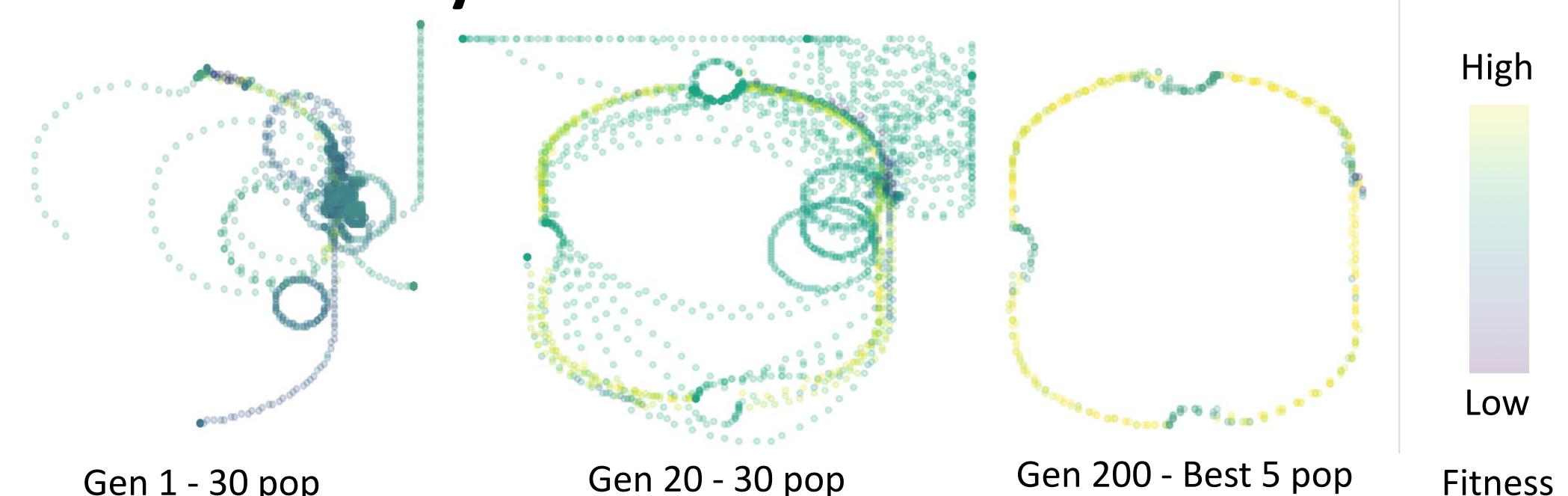


Fig 3.4 – Evolution process first to last generation, 30 populations

Fig 3.5 – Time taken for a whole circle

1:13 s 1:12 s 1:14 s

References

- Cyberotics (2019).** Cyberbotics: doc. [online] Cyberbotics.com
<https://cyberbotics.com/doc/guide/epuck?version=R2022b> [Accessed 14 Nov. 2025]
- Obstacle Avoidance and Trajectory Following Experiment with e-puck using Evolutionary Algorithms.**
https://www.academia.edu/19738739/Obstacle_Avoidance_and_Trajectory_Following_Experiment_with_e_puck_using_Evolutionary_Algorithms [Accessed 27 Nov. 2025]